

Proton Pass Heuristic Evaluation Report

Introduction

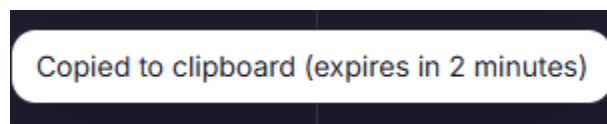
Proton Pass is a password manager developed by Proton AG alongside Proton Mail and Proton VPN. Password managers are designed to securely store passwords, logins, passkeys, and other sensitive information. Proton Pass is open source, uses end-to-end encryption, and markets itself as a secure way to store and manage passwords. With Proton Pass, users can store login credentials, generate secure passwords, create email aliases that can hide their email, manage identity information, and monitor accounts for potential breaches. It is available as a browser extension and a desktop application.

For our evaluation, we tested the free version on both Mac and Windows. We followed and used Nielsen's ten usability heuristics as our framework. Each group member independently walked through the app, creating and deleting logins, generating passwords, importing credentials, exploring settings, and testing autofill on real websites using the browser extension. After combining all our findings, we identified around 10 usability issues. These issues are a part of the four main themes, which are system feedback, confusing terminology, lack of error prevention, and navigation problems. Our severity ratings were using Nielsen's 0-4 scale, where 0 - not a usability problem, 1 - cosmetic problem, 2 - minor usability issue, 3 - major usability problem, 4 - usability catastrophe that must be fixed immediately.

Usability Issues

Visibility of System Status

One major theme across all the evaluations was the weak system feedback. In a security tool, users need to clearly understand what just happened. For example, when copying a password from the item detail view in the desktop or web app, a small notification appears at the bottom of the window indicating that the password has been saved to the clipboard. However, the notification goes away quickly and is easy to miss, especially on larger screens. Below is a screenshot of what this notification looks like.



The copy icon in the item detail panel does not visually change. Because passwords are sensitive, users need stronger confirmation before moving on. It is a minor usability issue, so the severity rating is 2.

Proton's item creation form permits saving weak passwords without any preventive measures. The create button remains active regardless of password strength, bypassing an opportunity to enforce secure behavior. Furthermore, the strength indicator provides no actionable guidance: it fails to communicate what specific criteria the password violates, such as insufficient length or missing character types. This leaves users without the information needed to make informed security decisions. For a security-focused product, the absence of clear password creation rules and meaningful feedback represents a gap in guiding users toward safer practices; hence, this usability issue was rated 2 on the severity scale.

Autofill failures also happen silently when using the browser extension on live login pages. If a website uses a non-standard login form, the extension pop-up would show no saved credentials. There is no message inside the extension panel explaining that the form may be incompatible. Users might assume they never saved the password and unnecessarily reset it. This was rated a severity 2 because it damages the trust in the tool.

Dark web monitoring is shown in the Pass Monitor section, which the user can get to through the shield icon in the left sidebar of the web app. However, the free tier users are not notified that monitoring is inactive until they enter the section. Discovering that a security feature is silently off compromises trust and creates a false sense of security; hence, this issue was rated a 2 on the severity rating.

Another issue is the browser extension toolbar icon in Chrome. The icon looks identical whether the vault is locked or unlocked. The only way to check the vault's lock status is to click the extension and open the pop-up panel. Since vault locking is important for security, not being able to see that status directly in the browser toolbar is risky. Leaving the vault unlocked may compromise sensitive information, which could lead to negative consequences. For this reason, we rated this a 3 on the severity scale.

Overall, Proton Pass doesn't always communicate system status, which is important in a password manager.

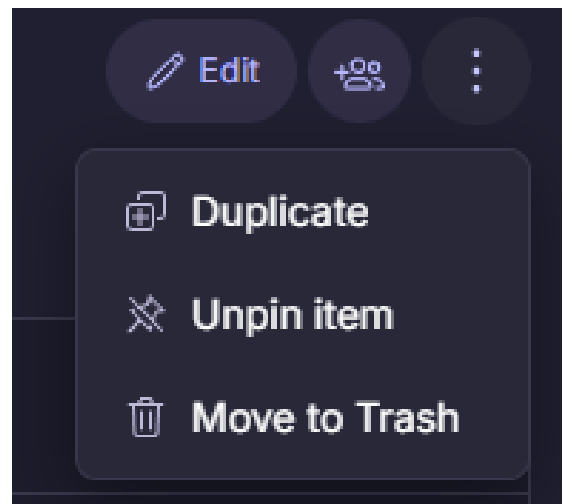
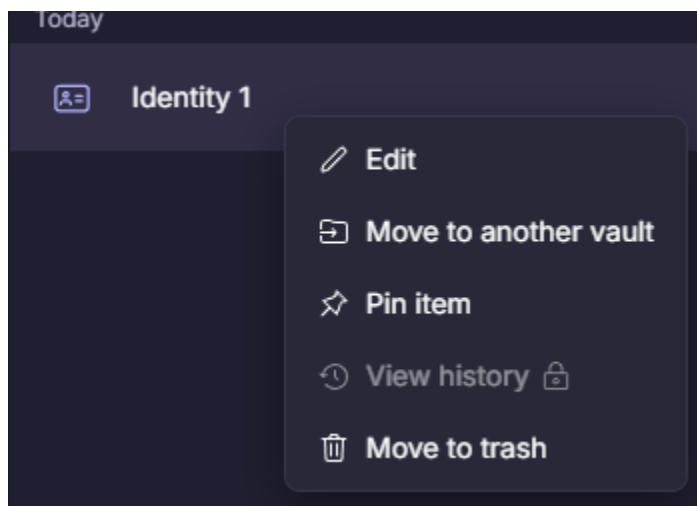
Consistency and Standards

Consistency issues in Proton Pass extend beyond individual features. Across the desktop app, browser extension, and web app, the interface does not always behave the same way or offer the same options. Some actions are available in one platform but not another, and there is no indication to the user that this is by design. Keyboard shortcuts further support this inconsistency: some actions support them while others require mouse navigation through menus, with no clear logic behind which interactions get shortcut support and which don't. Users who build habits in one part of the app are frequently caught off guard when those same habits don't transfer elsewhere. For a tool that people use repeatedly throughout the day, this unpredictability adds up. For this reason, it was rated a 3 on the severity scale.

Recognition rather than recall

Another pattern that we saw was terminology that doesn't match how an average user thinks. The word “vault” appears on the welcome screen and in the left-hand sidebar of the desktop and web app, where users see labels like personal vault and my vault. There are no explanations when hovering over the labels. This term may not be universally understood by less technical users who may assume that the passwords are simply saved, rather than stored in a container. This is a minor usability issue and was rated 2 on the severity scale.

Another usability issue we've identified is the lack of an option to duplicate an item when the user right-clicks on it from the Items view. In order to do so, the user would have to locate the “More Options” button in the top-right corner, which isn't immediately visible or recognizable. Below is a screenshot of what it looks like.



Since duplicating items is a common workflow, burying it behind a non-obvious button adds friction without reason. We rated this a severity 3.

Match between the system and the real world

When adding a new item to the vault on the item selection screen, users have to select a category. In Proton Pass, the category options map poorly to user intent. A user who wants to save an existing password for an account they already have must select "Login", while a user who wants to generate a brand new password must select "Password". These are not obviously distinct tasks to a new user, as both involve passwords, and the categorical distinction between them is not explained anywhere in the interface. This presents as a minor usability issue for certain users and could be resolved fairly quickly, so this issue was rated 2 on the severity scale.

The “hide my email” feature appears in the browser extension autofill dropdown on registration forms. It is shown next to the user's real email address without explanation. Someone who is unfamiliar with email aliases could select it without understanding that it creates a masked forwarding address. This could confuse when verification emails go to a different alias; hence, the issue was given a rating of 2.

Flexibility and Efficiency of Use

Proton Pass offers no keyboard shortcuts or advanced settings for experienced users. All actions require navigating through menus, which slows down repetitive tasks. This issue ties back to the broader point that Proton Pass does not behave consistently across its systems. This is a minor issue for general users but a noticeable friction point for experienced users, so it was rated 2 on the severity scale.

Error Prevention

Since Proton Pass is a security tool, preventing mistakes is important, and we found several weaknesses during the evaluation. There is no confirmation prompt shown when the delete button is pressed on a set of credentials. This can cause many problems, as if the button was pressed accidentally, or the information was not remembered, then it will be gone and difficult to recover. Although there is a method to recover deleted information, the fact that there is no confirmation prompt is concerning and should be implemented as soon as possible. It is a massive usability problem, so it was rated 4 on the severity scale.

Moving items between the vaults is done through the item options menu inside the item detail view. Once it is moved, there is no undo option. In shared vault citations, this could expose information to unintended users. Having sensitive information exposed could lead to disastrous consequences, making this usability issue a 4 on the severity scale.

Help and Documentation

Proton Pass has two separate settings panels: the browser extension settings panel, which is the gear icon inside the extension pop-up, and the full account settings page in the web app, which is accessed through the profile icon. There is no visible link between them: for example, for two-factor authentication, settings are only available in the web app account settings, not in the extension settings. A user searching within the extension might assume 2FA doesn't exist since 2FA is a major security feature; we rated this a severity 3.

The import screen, found under Settings in the app, only presents a file upload option without explaining how to first export passwords from another password manager. Users migrating from a different tool are left to figure out the export process on their own, which adds unnecessary friction to an already unfamiliar process. We rated this a severity 3.

Documentation and assistance are also significant problems. If the user would like assistance or information about Proton Pass, they must always be sent to a website or a video outside of the application. Not having guides or inline instructions, and instead relying on external resources to access information, inconveniences the user more, especially when the manager is initially very vague and visually unresponsive to some user input. This is a severity of 2 in isolation, but it would be worse if taken into account with the whole application.

Recommendations

Improve Feedback

Making sure the user receives a visible and informative response for the activities done in the manager would greatly improve the user experience for people. Make important notifications, such as when a password is copied, more prevalent, and make sure the user is aware when a vault is locked or unlocked. If a password autofill fails, make sure that the user is given some reason why.

Clarify Terminology & Onboarding

Make sure that users can understand how to effectively use the application and understand the terminology in the application without having to navigate through external sites: inline tutorials and in-app explanations would work very well in these cases.

Add Safe Guards for Risky Action

I believe that Proton Pass should also add stronger safeguards to prevent users from making mistakes that could cause real problems. Some actions in the app are irreversible; for example, deleting a login from the item detail view removes the credential without asking for confirmation. We believe that adding a confirmation dialog would prevent accidental deletions.

Also, if a user generates a password and tries to leave the page without copying it, the app should display a warning so that the password isn't lost. The system could also improve its security guidance by giving a better explanation when a password is seen as weak, and hopefully, it would encourage users to create a stronger one before saving it.

Strengthen Settings & Improve Help

Lastly, the app would benefit from better navigation and helpful in-app guidance. At the moment, Proton Pass has separate browser extension settings and full account settings in two different panels with no connection between them. If we were to combine these settings into one location, it would make the system easier to understand. The premium features, like the dark web monitoring, should also be labeled clearly before a user opens those sections so that free users know what they should expect. Also, proton pass could improve the help experience by having short explanations and guides that are in the interface instead of relying on outside resources. For example, the import screen should have instructions on how to export passwords from other password managers before uploading them. We believe these changes would make it easier for newer users to learn the system and find what they need.

Usability Issues Master List

0 = Not a Problem, 1 = Cosmetic, 2 = Minor Usability Issues, 3 = Major Usability Issues, 4, Usability Catastrophe

Problem	Severity (0-4)
Unclear system status and feedback (autofill notifications disappear quickly, export actions lack confirmation, copy password feedback is easy to miss, and unclear dark web monitoring status).	2
The potential of information being compromised due to the vault lock status not being visible in the browser toolbar.	3
Inconsistent behavior across platforms.	3
Confusing terminology (“vaults” naming convention).	2
Mismatch between the system and the real world (unclear distinction between certain item categories, confusing “hide my email” feature).	2
Limited workflow efficiency for advanced users due to a lack of keyboard shortcuts/advanced settings.	2

Poor deletion and undo mechanisms (no confirmation before deleting items, the undo option disappears quickly, hard to find the trash afterwards).	4
Poor discoverability and feature guidance (separate settings panels, hidden duplicate item option).	3
Secure sharing is buried in menus	2
No guide for the onboarding process	2
The import screen doesn't have any export instructions	3